

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology
Practical - 30

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Class: BSC IT	Batch:-----
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Aim: Cluster a dataset into two groups using k-means clustering in Breeze.

Code:

```
build.sbt x Demo30.scala
1 ThisBuild / version := "0.1.0-SNAPSHOT"
2
3 ThisBuild / scalaVersion := "2.13.12"
4
5 libraryDependencies += "org.scalanlp" %% "breeze" % "2.1.0"
6
7 lazy val root = (project in file("."))
8   .settings(
9     name := "Demo30"
10  )
11
```

```
1 import breeze.linalg._
2 object Demo30 extends App { new *
3
4   val data = DenseVector(1.0, 2.0, 3.0, 10.0, 11.0, 12.0)
5
6   // Initial cluster centers
7   var center1 = 2.0
8   var center2 = 10.0
9
10  // Repeat clustering
11  for (i <- 1 to 5) {
12    val cluster1 = data.toArray.filter(x =>
13      | math.abs(x - center1) < math.abs(x - center2)
14    )
15
16    val cluster2 = data.toArray.filter(x =>
17      | math.abs(x - center2) <= math.abs(x - center1)
18    )
19    // Calculate new centers
20    center1 = cluster1.sum / cluster1.length
21    center2 = cluster2.sum / cluster2.length
22  }
23  // Display clusters
24  println("Cluster 1:")
25  println(data.toArray.filter(x =>
26    | math.abs(x - center1) < math.abs(x - center2)
27  ).mkString(", "))
```

```
28
29     println("Cluster 2:")
30     println(data.toArray.filter(x =>
31         | math.abs(x - center2) <= math.abs(x - center1)
32     ).mkString(", "))
33
34     println("Center 1 = " + center1)
35     println("Center 2 = " + center2)
36
37     println("\nAbhishek_S046")
38 }
```

Output:

```
"C:\Program Files\Java\jdk1.8.0_341\bin\java.exe" ...
Cluster 1:
1.0, 2.0, 3.0
Cluster 2:
10.0, 11.0, 12.0
Center 1 = 2.0
Center 2 = 11.0

Abhishek_S046

Process finished with exit code 0
```